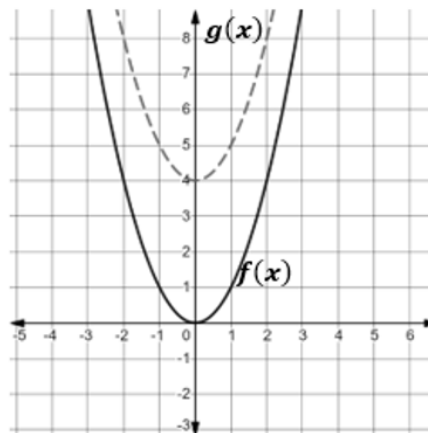


Algebra 1
Unit 13
Lesson 6 Practice

Name _____
Date _____
Period _____

The graphs of $y = f(x)$ and $y = g(x)$ are shown.



- Describe how to shift the graph of $f(x)$ so that it coincides with the graph of $g(x)$.
- Use the graph to complete the table of values for $f(x)$ and $g(x)$.

$f(x)$		$g(x)$	
x	y	x	y
-1		-1	
0		0	
1		1	

- Explain the change in the y -values from $f(x)$ to $g(x)$.

- Complete the statement:

The graph of $g(x)$ is a
☐ horizontal
☐ vertical
 shift of $f(x)$.

- Complete the statement:

The shift is seen in the table by
☐ adding
☐ subtracting
 4 to each
☐ x -value
☐ y -value

for each corresponding
☐ x -value.
☐ y -value.

For 6-7, complete the statements to describe how the graph of $g(x) = f(x) - 5$ compares to the graph of $f(x) = |x + 2|$.

6. The graph of $g(x)$ is a

horizontal
vertical

 shift of $f(x)$.

7. The graph of $g(x)$ is shifted _____ units

up from
down from
to the right of
to the left of

 the graph of $f(x)$.

8. Describe the effect on the x -values of $K(x) = |x - 3|$ for the transformation $J(x) = K(x) + 6$.

9. Describe the effect on the y -values of $v(x) = \frac{3}{4}x + 7$ for the transformation $w(x) = v(x) - 3$.

10. Describe how the graph of $T(x) = S(x - 10)$ compares to the graph of $S(x) = x^2 + 2$.